Full forms and meanings

Alphabetical reference to course terminology. Meanings state the relevant context; lesson links open the detailed daily PDFs.

A-C · D-F · G-L · M-P · Q-S · T-Z

SDC has distinct logic and timing meanings. CDC here is controllability don't-care. Liberty is a format name; tCQ, tSU and tH are symbols. Operation names and conventional abbreviations are identified by context.

A-C

ASIC	**Application-Specific Integrated Circuit** A chip designed for a particular application.	Lesson 02
AST	**Abstract Syntax Tree** Hierarchical representation of parsed source syntax.	Lesson 15
ATE	**Automatic Test Equipment** Applies and measures manufactured-device test patterns.	Lesson 08
ATPG	**Automatic Test Pattern Generation** Derives patterns to detect modeled manufacturing faults; fault-model coverage differs from functional coverage.	Explanation
BCP	**Boolean Constraint Propagation** Repeated deduction from unit clauses.	Lesson 23
BDD	**Binary Decision Diagram** Graph representing recursive Boolean decisions.	Lesson 21
BEOL	**Back End of Line** Fabrication of interconnect and related structures after the transistor/device stages; not the entire physical-design workflow.	Explanation
BIST	**Built-In Self-Test** On-chip pattern generation and response evaluation for a defined test target.	Explanation
BMC	**Bounded Model Checking** Checks unfolded behavior through a specified cycle bound.	Lesson 24
CCS	**Composite Current Source** More detailed current-based library model.	Lesson 26
CDC	**Controllability Don't-Care** Impossible internal input combination caused by network dependencies.	Lesson 19
CEC	**Combinational Equivalence Checking** Compares corresponding combinational functions.	Lesson 25
CMOS	**Complementary Metal-Oxide-Semiconductor** Uses complementary n-channel and p-channel devices; the inverter illustrates pull-down and pull-up behavior.	Lesson 01
CNF	**Conjunctive Normal Form** AND of clauses, each an OR of literals.	Lesson 23

CPU	**Central Processing Unit** Executes an instruction set; an architecture may allocate some behavior to software running on it.	Lesson 04
CTS	**Clock Tree Synthesis** Builds a clock-distribution network.	Explanation
CTS skew	**Capture arrival minus launch arrival, S=C−L** Positive pair skew adds setup margin and removes hold margin for the stated same-clock path.	Explanation

D-F

DC	**Don't-Care** Behavior permitted to vary under justified specification conditions.	Lesson 17
DEF	**Design Exchange Format** Exchanges a design instance's physical implementation data, such as placement, nets and routing.	Lesson 07
DFT	**Design For Test** Structures that improve manufactured-device testability.	Explanation
DPLL	**Davis-Putnam-Logemann-Loveland** SAT search through decisions, implications and backtracking.	Lesson 23
DRC	**Design Rule Checking** Checks geometric manufacturing rules.	Explanation
DUT	**Design Under Test** Implementation exercised by a verification environment.	Lesson 13
DVFS	**Dynamic Voltage and Frequency Scaling** Changes voltage and frequency together; distinguish power from energy for a fixed task.	Explanation
ECO	**Engineering Change Order** A controlled connectivity/geometry change followed by renewed affected verification.	Explanation
ECSM	**Effective Current Source Model** Current-source model introduced alongside NLDM and CCS.	Lesson 26
EDA	**Electronic Design Automation** Tools for design, implementation and verification.	Lesson 05
EM	**Electromigration** Current/stress-related material transport requiring technology-qualified reliability limits.	Explanation
EPI	**Essential Prime Implicant** Prime covering a required minterm uniquely among primes.	Lesson 17
ERC	**Electrical Rule Checking** Checks electrical connection and usage rules defined by the technology and rule deck.	Explanation

ESL	**Equivalent Series Inductance** A capacitor connection/model has inductance that limits rapid current delivery.	Explanation
ESR	**Equivalent Series Resistance** A capacitor model includes resistive loss and voltage drop in addition to ideal charge storage.	Explanation
FEOL	**Front End of Line** Fabrication stages that form device structures, before most interconnect processing.	Explanation
FPGA	**Field-Programmable Gate Array** Configurable logic and routing programmed after manufacture.	Lesson 02
FSM	**Finite State Machine** State, transition and output model for sequential behavior.	Lesson 20

G-L

GBA	**Graph-Based Analysis** Propagates timing bounds through graph vertices.	Lesson 29
GDSII	**Graphic Data System II** Layout exchange format used for geometric mask data.	Lesson 09
HDL	**Hardware Description Language** Models concurrent hardware behavior and structure.	Lesson 11
HLS	**High-Level Synthesis** Schedules and binds behavioral operations into hardware.	Lesson 14
HPWL	**Half-Perimeter Wire Length** Bounding-box estimate used in placement; does not specify a legal routed tree.	Explanation
IC	**Integrated Circuit** Devices and interconnect integrated on a substrate.	Lesson 01
IP	**Intellectual Property** Reusable design block and supporting integration information.	Lesson 04
LEF	**Library Exchange Format** Abstract physical information for cells and routing.	Lesson 26
LFSR	**Linear Feedback Shift Register** Deterministic shift/XOR recurrence; taps, convention and seed set its sequence.	Explanation
Liberty	**Liberty timing-library format** Cell functions, timing, units and power models in .lib files.	Lesson 26
LVS	**Layout Versus Schematic** Checks extracted layout connectivity against a design reference.	Explanation

M-P

MISR	**Multiple-Input Signature Register** Compacts several response bits per step into a finite signature, with possible aliasing.	Explanation
MMMC	**Multi-Mode Multi-Corner** Checks justified combinations of mode and timing conditions.	Lesson 30
MOSFET	**Metal-Oxide-Semiconductor Field-Effect Transistor** Gate voltage controls channel conduction; complementary device types form CMOS logic.	Lesson 01
NAND	**NOT-AND** Boolean complement of AND; its output is zero only when every input is one.	Lesson 17
NBA	**Nonblocking Assignment** Schedules a later left-side update after evaluating the right side.	Lesson 13
NLDM	**Nonlinear Delay Model** Lookup models of delay and transition versus slew and load.	Lesson 26
NMOS	**N-channel Metal-Oxide-Semiconductor** N-channel MOS device; the CMOS inverter uses it for the pull-down path.	Lesson 01
NOR	**NOT-OR** Boolean complement of OR; its output is one only when every input is zero.	Lesson 17
OCV	**On-Chip Variation** Models local timing variation using appropriate early/late effects.	Lesson 30
ODC	**Observability Don't-Care** Internal variation masked at all relevant observed outputs.	Lesson 19
OPC	**Optical Proximity Correction** Adjusts mask geometry to compensate for predictable lithographic printing distortion.	Lesson 09
PBA	**Path-Based Analysis** Recomputes selected paths with path-consistent quantities.	Lesson 29
PDK	**Process Design Kit** Process models, layers and design rules.	Lesson 01
PDN	**Power Delivery Network** Connected supply/ground structures whose impedance affects local voltage, timing and reliability.	Explanation
PI	**Prime Implicant** A legal product cube that cannot be further expanded.	Lesson 17
PMOS	**P-channel Metal-Oxide-Semiconductor** P-channel MOS device; the CMOS inverter uses it for the pull-up path.	Lesson 01
POS	**Product of Sums** AND of OR terms in Boolean notation.	Lesson 17

PPA	**Power, Performance, Area** Interacting implementation quality measures.	Lesson 02
PVT	**Process, Voltage, Temperature** Operating/characterization conditions of timing models.	Lesson 30

Q-S

RC	**Resistance-Capacitance** Interconnect resistance and capacitance contribute to delay, transition and loading.	Lesson 29
ROBDD	**Reduced Ordered Binary Decision Diagram** Canonical for one fixed variable order, with reduction rules applied.	Lesson 21
RPI	**Redundant Prime Implicant** In the essentials-first classification, a nonessential prime whose ON points are already covered by selected essentials.	A-13 cover selection
RTL	**Register Transfer Level** Describes transfers and transformations between state elements.	Lesson 15
SAT	**Boolean Satisfiability** Existence of an assignment satisfying a Boolean formula.	Lesson 23
SDC (logic)	**Satisfiability Don't-Care** Impossible relation between local inputs and their computed output.	Lesson 19
SDC (timing)	**Synopsys Design Constraints** Tcl-based design/timing requirements.	Lesson 32
SEC	**Sequential Equivalence Checking** Compares behavior over legal sequences and related initial states.	Lesson 25
SoC	**System on Chip** Integrates major system functions and reusable blocks on one integrated circuit.	Lesson 04
SOP	**Sum of Products** OR of AND terms in Boolean notation.	Lesson 17
SPEF	**Standard Parasitic Exchange Format** Extracted interconnect resistance and capacitance information.	Lesson 29
SPI	**Selective Prime Implicant** Remaining nonessential prime that can cover a still-uncovered ON point; select a sufficient residual cover.	A-14 worked chart
STA	**Static Timing Analysis** Analyzes timing paths with models and constraints.	Lesson 28

T-Z

TB	**Testbench** Stimulus, checking and supporting simulation logic.	Lesson 13

Tcl	**Tool Command Language** Command language used by many EDA interfaces.	Lesson 10
tCQ	**Clock-to-Q delay** Delay from the launch clock pin to changing Q.	Lesson 28
tH	**Hold time** Required data stability after a capture edge.	Lesson 28
tSU	**Setup time** Required data stability before a capture edge.	Lesson 28
UNSAT	**Unsatisfiable** No satisfying assignment under the encoded conditions.	Lesson 23
VCD	**Value Change Dump** Timestamped signal changes for waveform inspection.	Lesson 13
VLSI	**Very Large Scale Integration** Integration scale motivating a staged design flow.	Lesson 03
XOR	**Exclusive OR** For two inputs, the output is one when they differ; multiple-input XOR represents odd parity.	Lesson 17

Reference trail

Meanings are concise study summaries. Each lesson link retains the lecture frames, timestamps and supporting sources. ATPG/BIST have later treatment in Weeks 9-10; FEOL/BEOL have further treatment in Week 10.

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